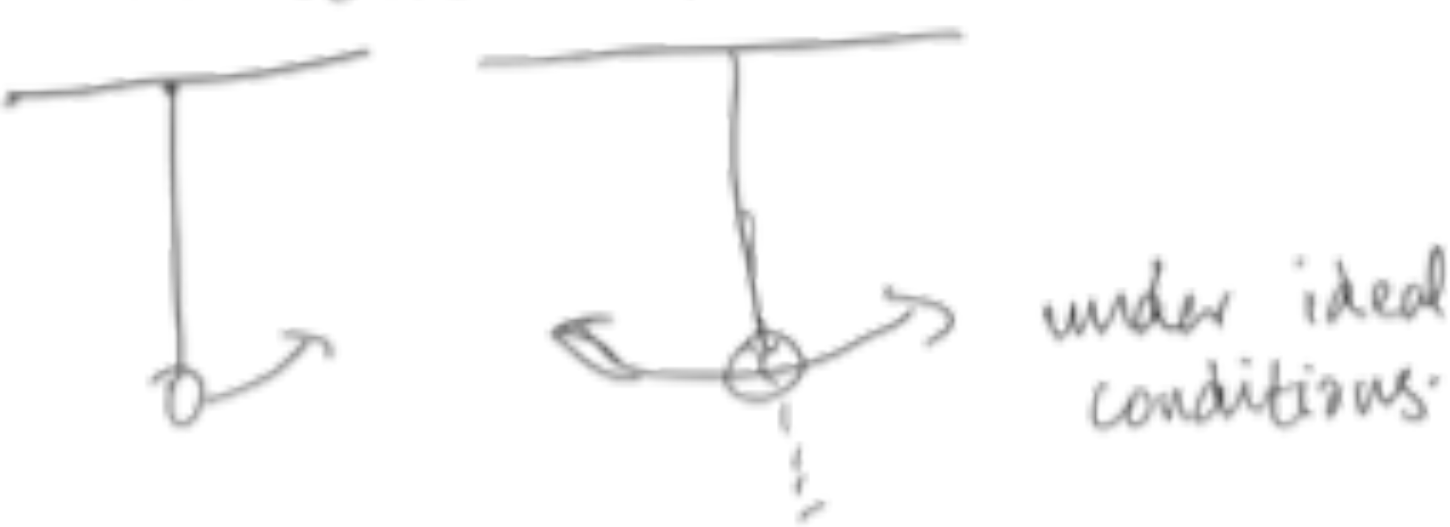
Postulates of Quantum Mechanics

1. Associated to any "isolated" physical systems, there is a complex vector space. The state of the system is described by the state vector which has unit norm.


under ideal conditions.

2. The evolution of a "closed" quantum system is described by a unitary transformation.

$$|\psi(t_2)\rangle = \underset{\sim}{\mathcal{U}}\,|\psi(t_1)\rangle$$

$$|\psi(t_2)\rangle = \mathcal{U}(t_2, t_1)\,|\psi(t_1)\rangle$$

2'. In the differential form. The time evolution of a closed quantum system is governed by the Schrodinger Eqn

$$i\hbar \frac{d|\psi\rangle}{dt} = H|\psi\rangle$$

where H is the Hamiltonian, which is hermitian.

$2' \rightarrow 2$

H is made up of quantum d.o.f.

The description of a system in terms of H is a bit tricky. Some assumptions are required.

H is a hermitian operator.

$$H = \sum_{\lambda} \lambda |\lambda\rangle\langle\lambda|$$

$|\lambda_0\rangle$ lowest eigenstate with lowest eigenvalue λ_0

$\underline{\text{ground state of the system}}$

λ : represents the energy of a particular state.

A single qubit $H = \hbar\omega X$ $\quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$|\lambda_+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|\lambda_-\rangle = \boxed{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}$$

$(+1, -1)$ $\quad -1$ is the ground state energy

$$|\lambda_-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$|\psi(t_2)\rangle = \exp\left(\frac{iH}{\hbar}(t_2 - t_1)\right)|\psi(t_1)\rangle$$

$$\mathcal{U}(t_2, t_1) = \exp\left(\frac{iH}{\hbar}(t_2 - t_1)\right)$$

A generic unitary operator $\mathcal{U} = \exp(iK)$ where K is hermitian.

$[A, B] = 0.$

$$\exp(A)\exp(B) = \exp(A + B)$$

This a consequence of the "Baker-Campbell-Hausdorff theorem."

$[A, B] \neq 0.$

$$\mathcal{U}(t_2, t_1) = \exp\left(i\frac{H}{\hbar}(t_2 - t_1)\right)$$

$$= \exp(i\alpha H)$$

$$H = \sum_{i=1}^{n} H_i = \underline{H_1 + H_2 + \cdots\ H_n}$$

$$[H_i, H_j] \neq 0$$

$$\exp(i\alpha H) = \exp\left(i\alpha \sum_i H_i\right)$$

$$\approx \exp(i\alpha H_1)\exp(i\alpha H_2) \cdots \exp(i\alpha H_n) + O(\alpha^2)\exp\left(\text{commutator of } H_i, H_j\right)$$

Suzuki-Trotter decomposition.

Do we have ideal quantum systems?

When we apply a unitary transformation

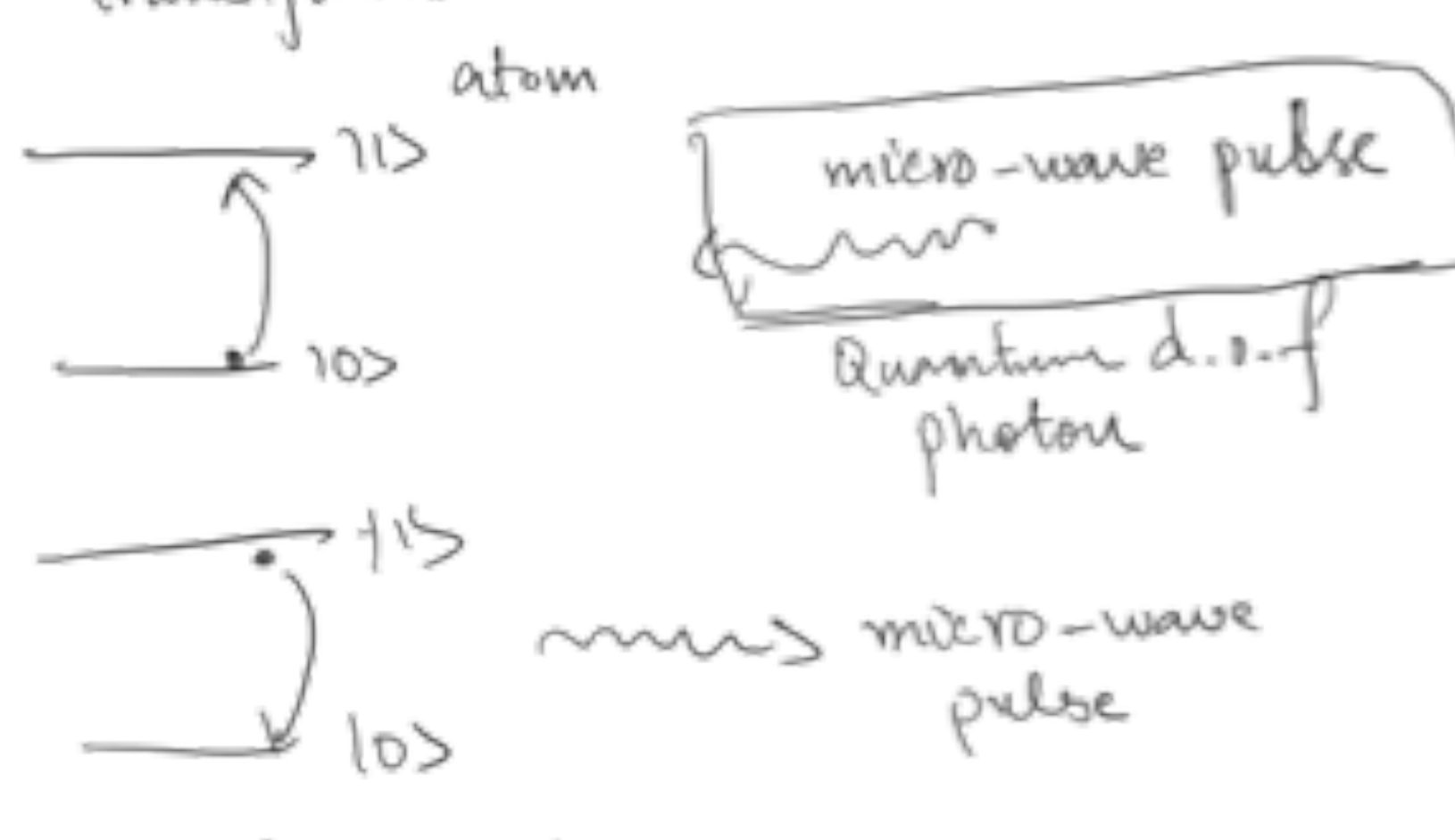
$$\underline{\mathcal{U}|\psi\rangle \rightarrow |\psi'\rangle}$$

$\mathcal{U}$ is implemented via a set of basic operations. "Gates" → Quantum gates

"we" interacting with the quantum system.

If we interact with the quantum system, it is no more closed or isolated.

The dynamics are no more unitary transformations.



laser-atom system.

$$(H_{atom} + H_{laser}) = H_{total}$$

We want to ignore the dynamics of the laser system.

$$V_{total} = V_{atom} \otimes V_{laser}$$

$\underline{\text{Atomic hamiltonian}}$ = just describes the dynamics of the atom.

The laser system is in the control of the experimentalist. and H_{total} is a time-varying operator.

We can say that the dynamics of the system are unitary.

If we interact with the system, we change it's quantum state.

Exception to the unitary evolution:

Yes, the quantum measurements are an exception.

Unitary evolution is reversible

$$\mathcal{U}|\psi\rangle \rightarrow |\psi'\rangle$$

$$\mathcal{U}^\dagger|\psi'\rangle \rightarrow |\psi\rangle$$

Quantum measurements are interactions with the system

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad |\psi\rangle =$$

$|0\rangle \leftarrow P = \frac{1}{2}$ $\qquad P(\underline{|0\rangle}) = \frac{1}{2}$

$|1\rangle \leftarrow P = \frac{1}{2}$ $\qquad P(\underline{|1\rangle}) = \frac{1}{2}.$

$|\psi\rangle = a|0\rangle + |1\rangle$ with $|a|^2 + |b|^2 = 1$

$\underline{|a|^2}$ represents probability of measuring the state $|0\rangle$

$\underline{|b|^2}$ . . . . $\qquad |1\rangle$

Postulate 3. Quantum measurements are described by a collection of measurement operators $\{M_m\}$ where m is the index of the outcome.

$|\psi\rangle$

$$p(m) = \underline{\langle\psi| M_m^\dagger M_m |\psi\rangle}$$

Once you measure the system, the state of the system collapses

Post-measurement state.

$$|\psi'\rangle = \frac{M_m|\psi\rangle}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$$

$$\sum_m M_m^\dagger M_m = I$$

⇒ statement that the sum of probabilities of all possible outcomes is 1

$$\sum_m p(m) = \underline{1}$$

$$\sum_m \langle\psi| M_m^\dagger M_m |\psi\rangle$$

$$\langle\psi| \underbrace{\sum M_m^\dagger M_m}_{} |\psi\rangle = 1$$

$$\Rightarrow \sum_m M_m^\dagger M_m = I$$

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measurement of a qubit in computational basis.

$M_0 = |0\rangle\langle 0|$ $\qquad M_0^2 = M_0$

$M_1 = |1\rangle\langle 1|$ $\qquad M_1^2 = M_1$

$I = M_0^\dagger M_0 + M_1^\dagger M_1$ $\qquad |\psi\rangle = a|0\rangle + b|1\rangle$

$I = |0\rangle\langle 0| + |1\rangle\langle 1|$

The probability of measuring $|0\rangle$ $m = 0$

$$p(0) = \langle\psi| M_0^\dagger M_0 |\psi\rangle = |a|^2$$

$$p(1) = \langle\psi| M_1^\dagger M_1 |\psi\rangle = |b|^2$$

$$\left.\begin{aligned} \frac{M_0|\psi\rangle}{|a|} &= \left(\frac{a}{|a|}\right)|0\rangle \\ \frac{M_1|\psi\rangle}{|b|} &= \left(\frac{b}{|b|}\right)|1\rangle \end{aligned}\right\}$$

The original state $|\psi\rangle = a|0\rangle + b|1\rangle$

Super-position of basis state.

Now recovering the original state $|\psi\rangle$ is impossible. ⇒ Measurement is a non-unitary process.

N-qubit system a single measurement will give you 2^N possible outcomes

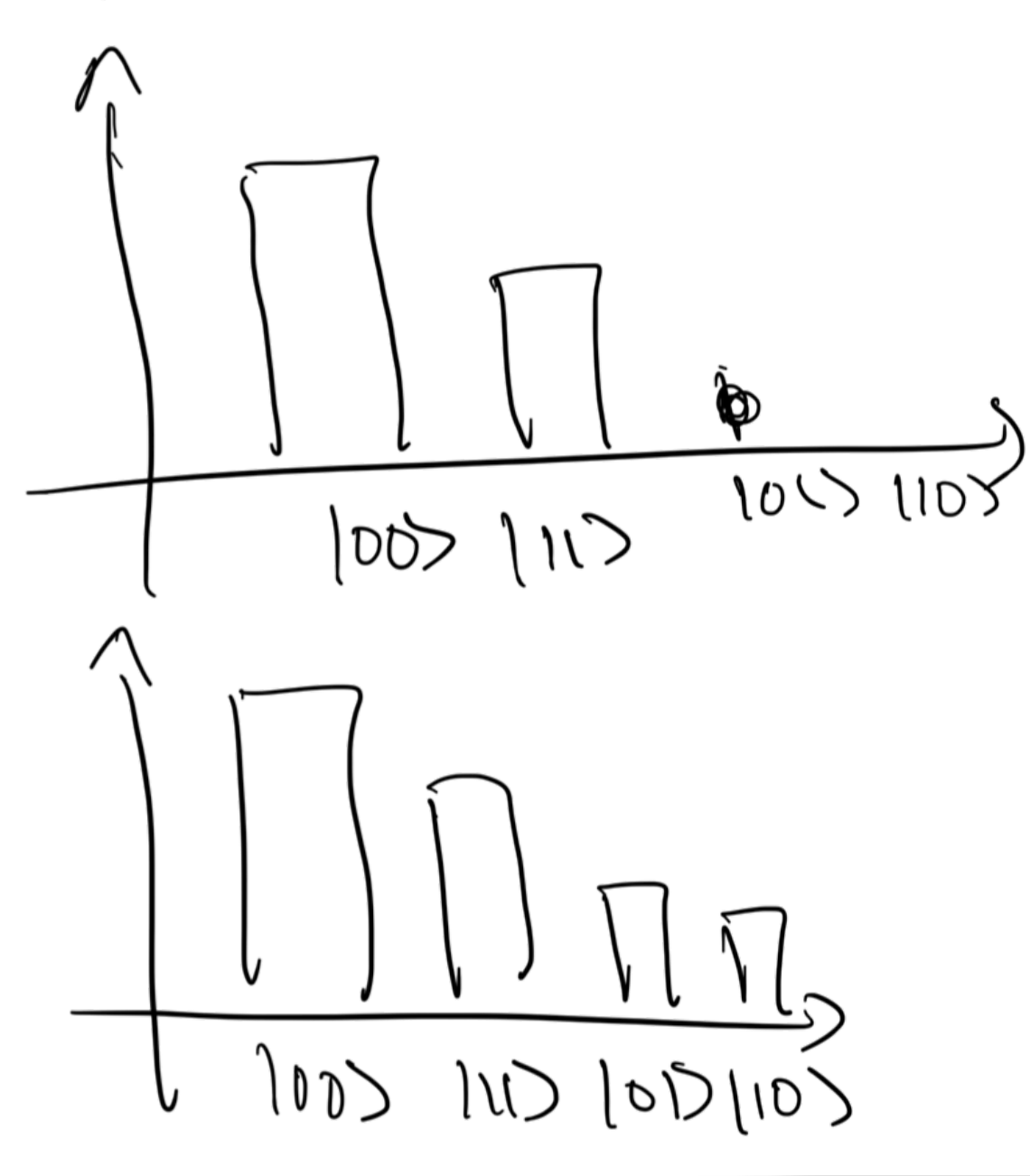
$|\psi_N\rangle \rightarrow \underline{10{,}000 \text{ measurements}}$

snapshots.

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \qquad \begin{matrix} |00\rangle & |01\rangle \\ |10\rangle & |11\rangle \end{matrix}$$

$$p(|00\rangle) = 5678/10{,}000$$

$$p(|11\rangle) = \frac{10{,}000 - 5678}{10{,}000}$$



Distinguish two quantum systems.

$|\psi\rangle$ $\quad |\varphi\rangle$

$$\langle\psi|\varphi\rangle = 0$$

⇒ They are distinguishable.

$$\langle\psi|\varphi\rangle \neq 0$$

Then there is no scheme of measurement that can distinguish between $|\psi\rangle$ and $|\varphi\rangle$

Scheme of measurement: Specifying measurement operators M_m

$$\langle\psi|\varphi\rangle = 0$$

$\underline{M_m^{\varphi}}$ $\qquad \underline{M_m^{\psi}}$ $\qquad$ Section 2.2.4

Projective measurements:

M, hermitian

$$M = \sum_m m P_m \qquad P_m = |m\rangle\langle m|$$

projection onto the eigenspace of M with eigenvalue m

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